

Frekvencia - átviteli függvények megjelenítése

→ Bode diagram

- amplitúdóarány (teljesítményarány) - körfrekvencia

$$A_r = \frac{A_{ki}}{A_{be}} \rightarrow P_r = \frac{P_{ki}}{P_{be}} = \frac{A_{ki}^2}{A_{be}^2} = A_r^2 [-]$$

$$B = \lg(P_r) [B] = 10 \lg(P_r) [dB] = 10 \lg(A_r^2)$$

↳ be

↳ ki/be

$$B = 20 \lg(A_r) [dB]$$

→ ezt jelenítjük meg a diagrammon

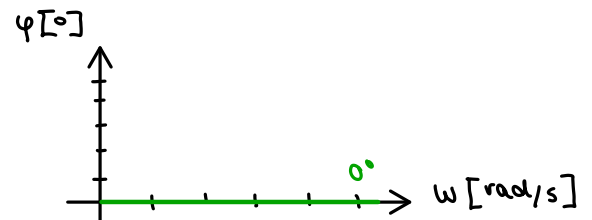
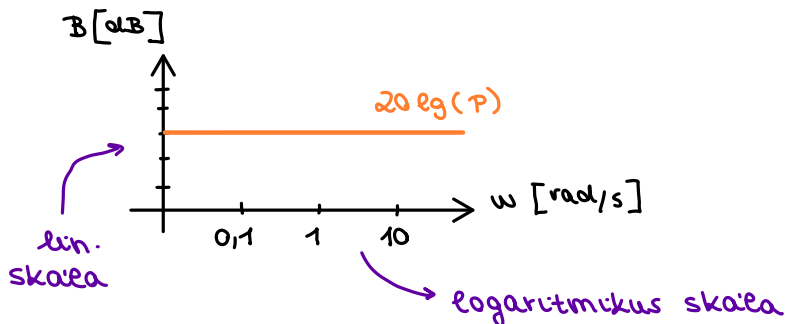
A_r -t hogy kapjuk meg? $|W(j\omega)| = A_r$ $\rightarrow j\omega$ FORMÁLIS behelyettesítés

- fázisszög (fáziskülönbség) - körfrekvencia

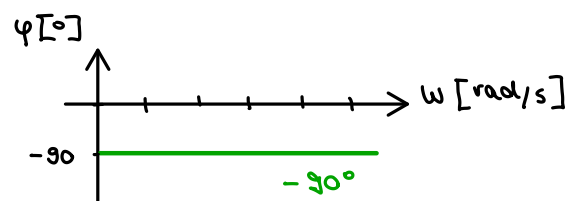
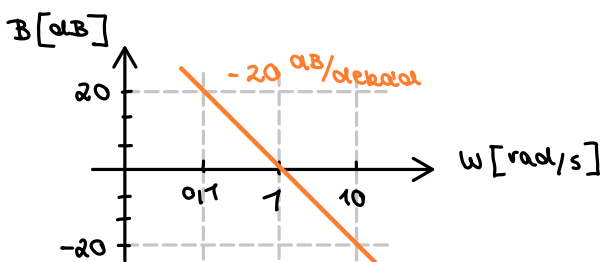
$$\angle W(j\omega) = \arg(W(j\omega)) = \Delta\varphi = \varphi_{ki} - \varphi_{be} \rightarrow \varphi [^\circ]$$

Ísmétlés (első. EA)

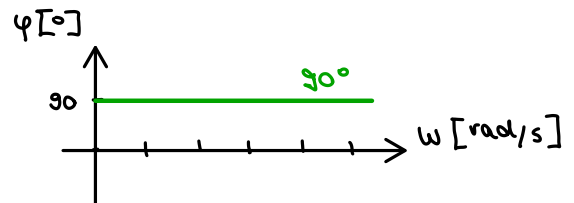
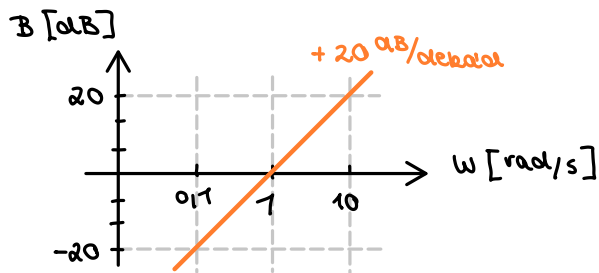
1.) arányos tag: $W(s) = P$



2.) integráld tag: $W(s) = \frac{1}{s}$ (ideális)



3.) Deriváló tag: $W(s) = s$ (ideális)

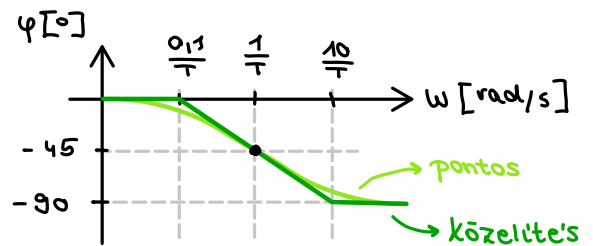
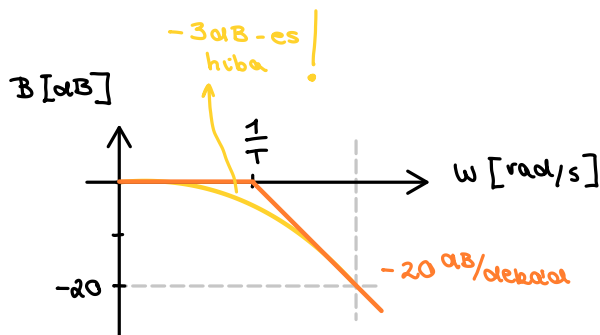


integráló tag tükrözése az 'x' tengelyre (reciproka)

4.) Tiszta pólus: $W(s) = \frac{1}{1+sT}$

↳ adott töréspontig ($1/T$) nem erősít, utána -20 dB/decade

↳ fázis: drótvékus $1/T$ frekvencia körüli közelítés
1-1 dekadálal jobbra és balra (lineárisan), többi helyen konstans

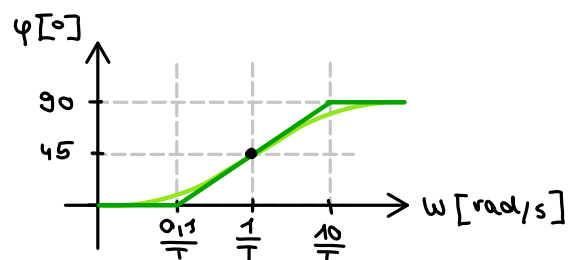
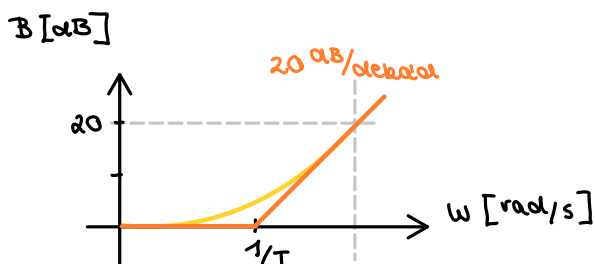


5.) Tiszta zérus: $W(s) = 1+sT$

↳ adott töréspontig ($1/T$) nem erősít, utána $+20$ dB/decade

↳ fázis: drótvékus $1/T$ frekvencia körüli közelítés
1-1 dekadálal jobbra és balra (lineárisan), többi helyen konstans

tiszta zérushoz hasonlóan



tiszta pólus tükrözése az 'x' tengelyre (reciproka)

① $W(s) = \frac{10}{s}$

a) $W(s) \rightarrow W(j\omega)$

$s = j\omega$
formális
behelyettesítés

$-j\omega$: 'j' konjugált párja

$$W(j\omega) = \frac{10}{j\omega} = \frac{10}{j\omega} \cdot \frac{-j\omega}{-j\omega} = -\frac{10j}{\omega}$$

$$\operatorname{Re}\{z\} = 0$$

$$\operatorname{Im}\{z\} = -\frac{10}{\omega}$$

↓

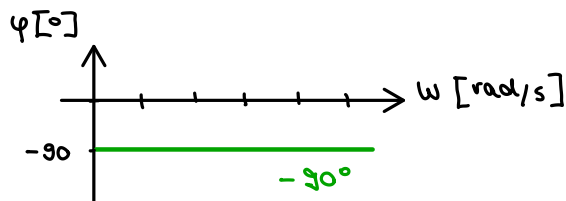
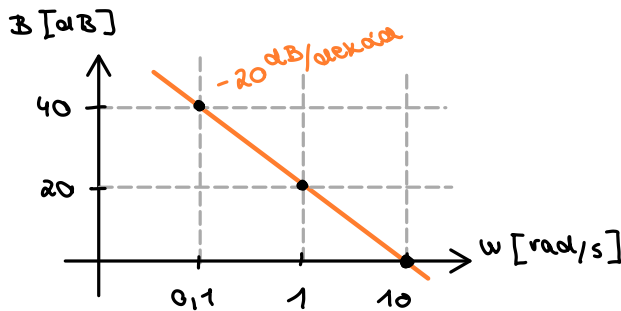
$$A_r = |W(j\omega)| = \sqrt{\operatorname{Re}^2 + \operatorname{Im}^2} =$$

$$= \sqrt{0^2 + \left(-\frac{10}{\omega}\right)^2} = \frac{10}{\omega}$$

$$B = 20 \lg\left(\frac{10}{\omega}\right) = \underbrace{20 \lg(10)}_{20} - 20 \lg(\omega)$$

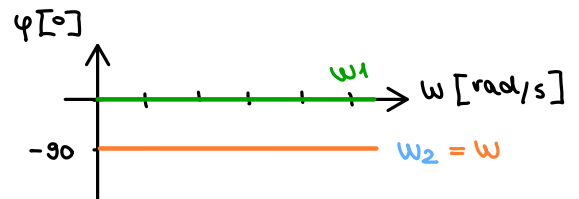
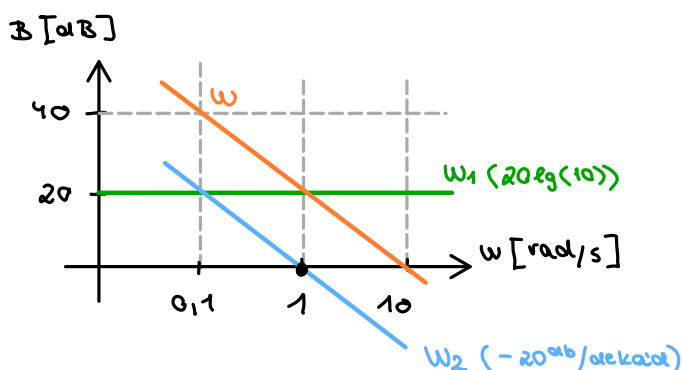
$$\operatorname{tg} \varphi = \frac{\operatorname{Im}\{z\}}{\operatorname{Re}\{z\}} = \frac{-10/\omega}{0}$$

$$\varphi = -90^\circ$$



b) egymással sorbakapcsolt alapelemek segítségével

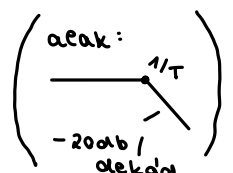
$$W(s) = \underbrace{10}_{\text{arányos } W_1(s)} \cdot \underbrace{\left(\frac{1}{s}\right)}_{\text{integrált } W_2(s)}$$

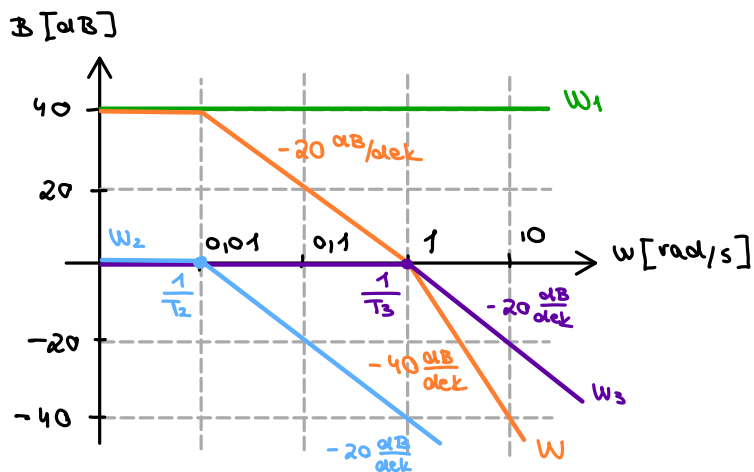


② $W(s) = \frac{100}{(1+100s)(1+s)} = \underbrace{100}_{\text{arányos } W_1} \cdot \underbrace{\frac{1}{1+100s}}_{\text{pólus } W_2} \cdot \underbrace{\frac{1}{1+s}}_{\text{pólus } W_3}$

→ töréspont: $W_2 \rightarrow \frac{1}{T_2} = \frac{1}{100}$

$W_3 \rightarrow \frac{1}{T_3} = 1$

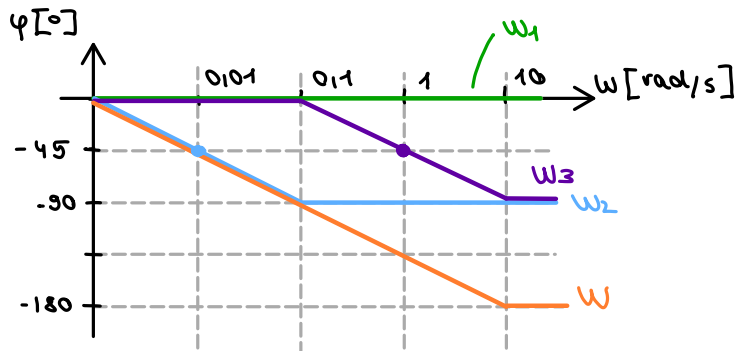




$$w_1 \rightarrow 20 \log(100) = 40$$

$$w_2 \rightarrow 1/T_2 - \log 0, \text{ utána } -20 \text{ dB/decad}$$

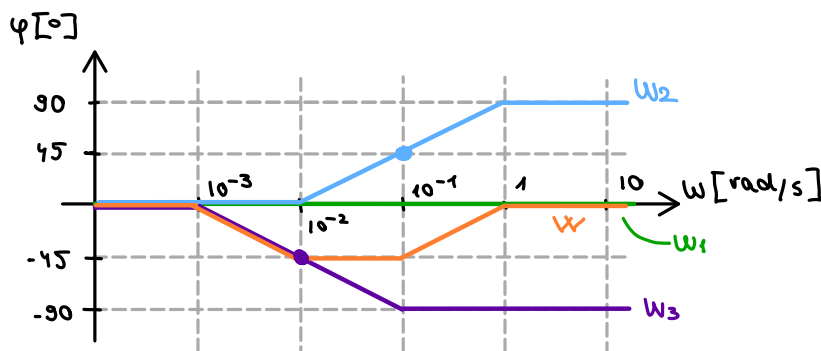
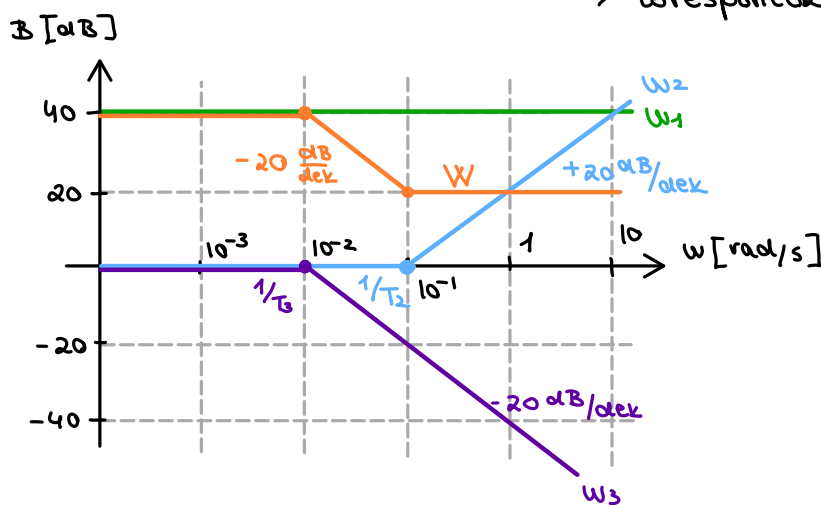
$$w_3 \rightarrow 1/T_3 - \log 0, \text{ utána } -20 \text{ dB/decad}$$



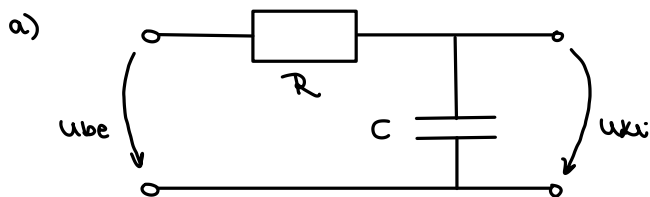
$$\textcircled{3} \quad W(s) = 100 \cdot \frac{1+10s}{1+100s} = \underbrace{100}_{w_1 \text{ arányos}} \cdot \underbrace{(1+10s)}_{w_2 \text{ zérus}} \cdot \underbrace{\frac{1}{1+100s}}_{w_3 \text{ pólus}}$$

$$\Rightarrow \text{töréspontok: } w_2 \rightarrow \frac{1}{T_2} = \frac{1}{10} \quad (10^{-1})$$

$$w_3 \rightarrow \frac{1}{T_3} = \frac{1}{100} \quad (10^{-2})$$

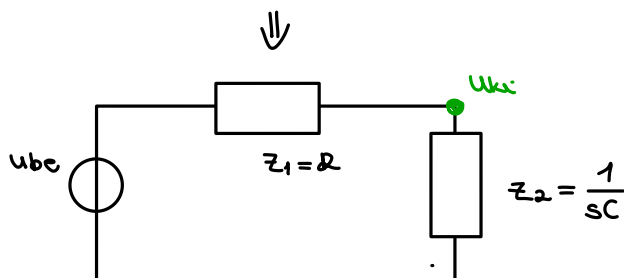


④ Átültetési szűrő



$$R = 10 \text{ k}\Omega = 10^4 \Omega$$

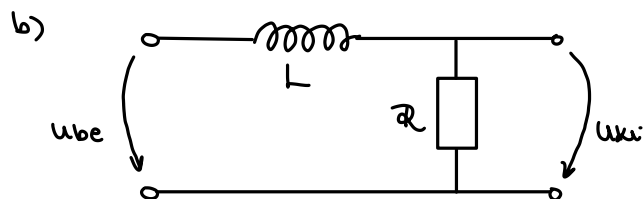
$$C = 10 \mu\text{F} = 10^{-5} \text{ F}$$



$$u_{ki} = \frac{Z_2}{Z_1 + Z_2} = \frac{1}{1 + \frac{Z_1}{Z_2}} u_{be}$$

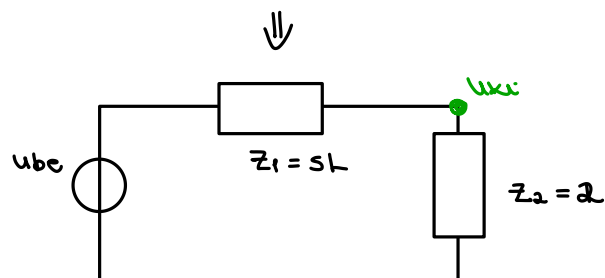
$$= \frac{1}{1 + RCs}$$

$T = RC = 0,1 \text{ s}$



$$L = 10^{-3} \text{ H}$$

$$R = 10^{-2} \Omega$$

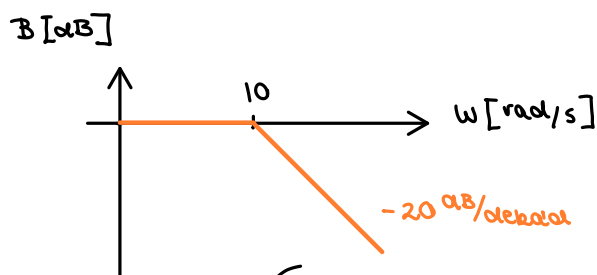


$$u_{ki} = \frac{Z_2}{Z_1 + Z_2} = \frac{1}{1 + \frac{Z_1}{Z_2}} u_{be}$$

$$= \frac{1}{1 + L/Rs}$$

$T = L/R = 0,1 \text{ s}$

Boole - diagram ugyanúgy néz ki
→ mint egy tiszta pótlás!

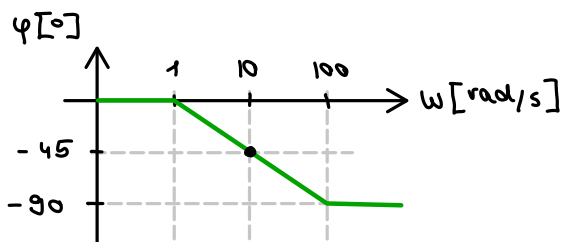


$B = 20 \log(A_r)$

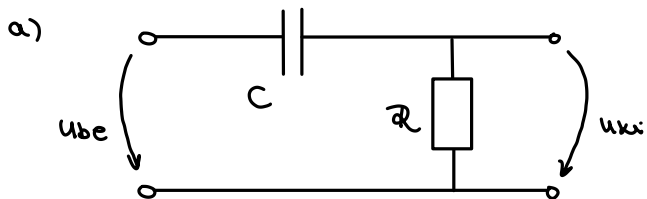
$\hookrightarrow A_r = \frac{A_{ki}}{A_{be}}$

→ ha B csökken → A_r is → $A_{ki} = A_r \cdot A_{be}$

azonos bemenetre
alatt frekvencián
kisebb a kimenet
amplitúdója



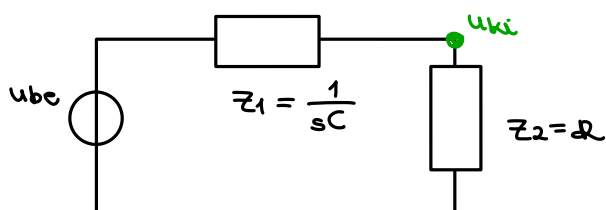
⑤ felüledáteresztő szűrő



$$R = 10 \Omega$$

$$C = 10 \mu F = 10^{-5} F$$

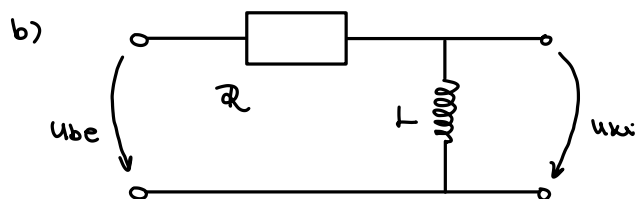
$\Downarrow$



$$u_{ki} = \frac{Z_2}{Z_1 + Z_2} = \frac{1}{1 + \frac{Z_1}{Z_2}} u_{be}$$

$$= \frac{RCs}{1 + RCs}$$

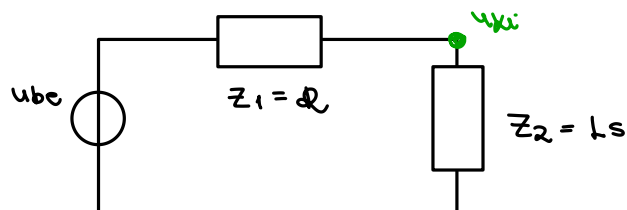
$T = 10^{-4} s$



$$L = 10^{-3} H$$

$$R = 10 \Omega$$

$\Downarrow$



$$u_{ki} = \frac{Z_2}{Z_1 + Z_2} = \frac{1}{1 + \frac{Z_1}{Z_2}} u_{be}$$

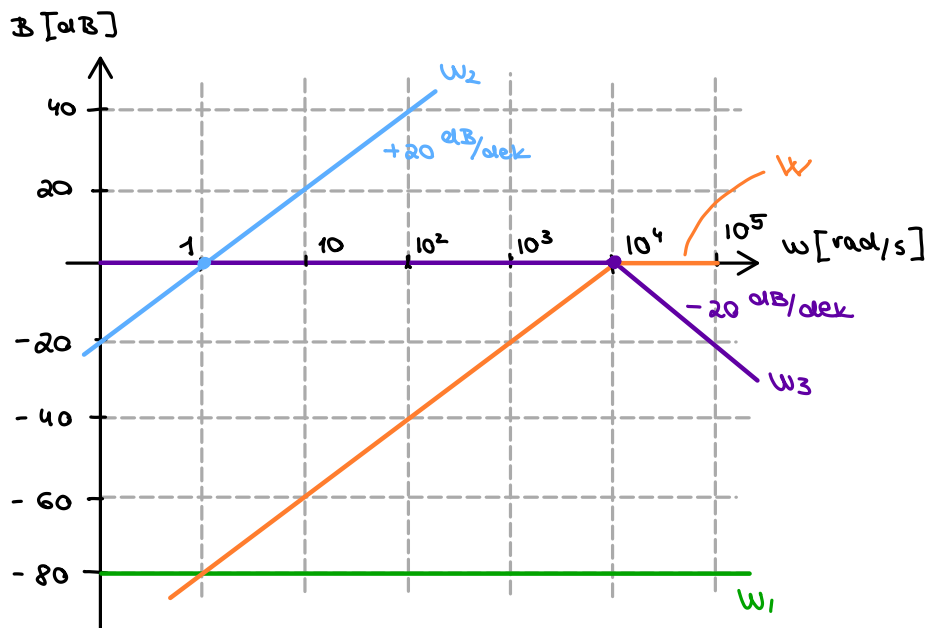
$$= \frac{L/Rs}{1 + L/Rs}$$

$T = 10^{-4} s$

azonos Bode

$$\rightarrow W = \frac{Ts}{1 + sT} = T \cdot s \cdot \frac{1}{1 + sT}$$

arányos derivált pólus



$$W_1 \rightarrow 20 \log(10^{-4}) = -80$$

